Xiyuan GUO

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EDUCATION

University of Virginia Charlottesville, VA — Sept 2024 – May 2027 (Expected)

Degree: Bachelor of Arts in Computer Science and Physics — GPA: 3.89/4.00

Independent Study: Quantum Information (Caltech Ph219A Curriculum), Adv. Quantum Mechanics (Sakurai)

PROFESSIONAL EXPERIENCE

Optical Physics Assistant, Yale (Liu's Group) New Haven, CT — Feb 2026 – Present

- Developing a constrained ADMM algorithmic framework for blind computational polarimetry. Evaluating ongoing reconstructions of the full Stokes polarization cube (S_0, S_1, S_2, S_3) from a single optical diffuser speckle measurement.
- Bypassing an infinitely underdetermined bottleneck utilizing Tikhonov warm-starts. Applying the Woodbury matrix identity to analytically reduce massive Gram matrix inversions into element-wise divisions using power spectra sums.
- Integrating strict physical constraints within optimization loops executing exact Lorentz cone orthogonal projections. This mathematically bounds recovered polarization components by total intensity, ensuring viable states.
- Mitigating frequency-domain cross-talk via intensity-weighted group soft-thresholding. Calibrating static optical system operators using spatial Gram-Schmidt orthogonalization to establish orthogonal bases, preserving power ratios.

Network Science Intern, UVA Biocomplexity Institute Charlottesville, VA — May 2026 – Present

- Formulated massive-scale network simulations modeling stochastic viral pathogen spread, implementing SEIR compartmental cascade models to capture heterogeneous transmission dynamics.
- Engineered attention-based Graph Neural Network (GNN) architectures predicting multi-hop epidemic propagation, leveraging spatiotemporal message-passing frameworks to extract latent topological features from dynamic contact networks.
- Collaborated within a transdisciplinary initiative to scale core epidemiological algorithms, bridging high-performance Bayesian data science with empirical public health parameters to inform global disaster resilience.

Research Software Engineer, UVA (Ke's Group) Charlottesville, VA — Oct 2025 – Present

- Architected a highly available web database and full-stack research platform on RHEL virtual machines, backed by DOE-BES and DOE-ARPA-E funding to drive frontier rare-earth materials research.
- Integrated massive-scale Crystallographic Information File (CIF) data with complex Density Functional Theory (DFT) and Dynamical Mean-Field Theory (DMFT) quantum mechanics calculations.
- Engineered cloud-native machine learning pipelines utilizing Go and Next.js to actively predict key material properties, including magnetic anisotropy, magnetic moments, Curie temperature, and crystallization.
- Created immersive browser-based visualizations using Three.js and JSmol, generating interactive 3D atomic models for the real-time structural inspection of theoretical computational outputs.

Quantitative Researcher, UVA (BAL Lab) Charlottesville, VA — May 2025 – Present

- Engineered a mathematical framework for multivariate longitudinal missing data imputation enforcing the Law of Total Variation, mitigating severe structural covariance bias in unconstrained predictive models.
- Formulated a Lagrange-constrained C++17 gradient descent engine, deriving custom matrix calculus gradients to project non-parametric estimates onto a smooth Riemannian positive-semidefinite (PSD) manifold.
- Executed comprehensive simulation studies comparing advanced deep generative imputation architectures—including score-based diffusion (CSDI), adversarial networks (GAIN), and transformers (SAITS)—against traditional FIML and TSRE baselines.
- Bounded highly non-convex search optimization spaces using robust statistical estimators, scaling dense matrix operations across Slurm HPC clusters for massive Monte Carlo simulations.

AWARDS & HONORS

- **Double Hoo Research Award:** Secured competitive grant funding to drive independent, interdisciplinary computational research.
- **Academic Honors:** USA Computing Olympiad (USACO) Silver Division; UVA Dean's List of Distinguished Students.
- **Open-Source Recognition:** Achieved Hacker News Front Page (#5, #6), Homebrew Core inclusion, and 342+ GitHub stars for `difi`.

CERTIFICATIONS & SKILLS

- **Math & Science:** Quantum Mechanics, Quantum Error Correction, Optics, Density Functional Theory Tensor Algebra, Topology
- **Machine Learning & AI:** PyTorch, TensorFlow, Keras, JAX, Scikit-Learn, XGBoost, Pandas, NumPy, OpenCV, RcppArmadillo
- **Infrastructure & Tools:** Linux (RHEL/Fedora), AWS (EC2, S3), Docker, Kubernetes, CUDA, Slurm HPC, Git, CI/CD, Vim, Emacs
- **Frameworks & Web:** React, Next.js, SvelteKit, Node.js, FastAPI, Bubble Tea, GraphQL, REST APIs, TailwindCSS
- **Languages:** Python, C/C++, Rust, Go, TypeScript, JavaScript, R, Julia, SQL, Lua, Bash, Java, LaTeX
- **Certifications:** AWS Certified AI Practitioner, AWS Certified Cloud Practitioner